

A Modern Theory of Mass Communication

Attention, Reach, and Belief Transmission in Economies with Heterogeneous Agents

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Abstract

We develop a theory of mass communication for economies populated by wealth-heterogeneous agents in the tradition of heterogeneous-agent New Keynesian (HANK) macroeconomics. Because agents differ in liquidity, stakes, and the opportunity cost of attention, a single broadcast message is decoded unevenly across the population: attention is rationally allocated, not uniformly given. We show that (i) discrete wealth tiers generate an endogenous, cost-dependent audience-selection problem for any sender; (ii) under linear payoffs this selection is bimodal (narrow-niche or full-cascade reach), a fragility resolved once Bayesian signal extraction is imposed; (iii) receivers facing genuine state uncertainty solve a sequential (Wald-type) attention-stopping problem that staggers aggregate responses across tiers; (iv) a sender who also controls a reach-free instrument (e.g., a central bank’s policy rate or balance sheet) optimally concentrates communication where the instrument itself does not already deliver value, generating a bounded, non-explosive complementarity between action and communication; and (v) the sender’s optimal precision path, solved in closed form via a Hamilton–Jacobi–Bellman equation, is an exact piecewise-linear “sawtooth” in cumulative evidence. We derive the associated welfare wedge, characterize efficient subsidy instruments, and discuss cross-disciplinary connections spanning statistics, physics, biology, psychology, political science, and computer science, together with policy implications for central bank communication and the regulation of mass media.

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1 Introduction

Classical theories of mass communication – from Shannon’s channel-capacity formalism to representative-agent macroeconomic models of “forward guidance” – implicitly treat the audience as a single receiver, or as a population whose heterogeneity is payoff-irrelevant. Once agents differ in wealth, liquidity, and marginal propensity to consume (MPC), as in the heterogeneous-agent New Keynesian (HANK) literature [1], this simplification breaks down: the *same* broadcast signal is optimally attended to by some agents and rationally ignored by others, and the aggregate consequence of a message depends on the joint distribution of stakes and attention costs, not merely on its informational content.

This paper builds a theory of mass communication on HANK foundations. The heterogeneity that matters is economic (wealth/liquidity, hence MPC and stake in different classes of news), not merely informational. We proceed in stages: static attention allocation across discrete wealth tiers; a sender’s reach-cost problem and its closed-form solution; the resolution of a cascade instability once attention is Bayesian rather than reduced-form; a genuine sequential-sampling (optimal stopping) microfoundation for *when* agents update; a dynamic control problem for the sender’s optimal precision path; the special case of a central bank wielding both a reach-free policy instrument and communication; and a full welfare and policy analysis.

2 Related Concepts and Cross-Disciplinary Foundations

The model draws on, and connects to, several literatures and formal traditions:

Economics/Macro-Finance: Heterogeneous-agent New Keynesian models [1], rational inattention [2], and the forward-guidance puzzle [3, 4].

Statistics: Sequential hypothesis testing and the sequential probability ratio test (SPRT) [5].

Physics: The evolution of cumulative evidence in our sequential-sampling model is a biased random walk, formally identical to a drift-diffusion process used in statistical mechanics and neuroscience of decision-making; posterior variance contraction is the informational analogue of entropy reduction.

Chemistry: The attention threshold ψ_k that must be cleared before a tier engages with a signal is structurally analogous to an activation energy in Arrhenius kinetics: below a threshold no reaction (attention) occurs, and the “rate” of engagement is sharply threshold-dependent rather than smooth.

Biology: The cascade/bimodality result (Proposition 1) is the communication-economics analogue of threshold epidemic dynamics (e.g., R_0 -driven contagion switches between fade-out and outbreak); the tier-by-tier staggered convergence in Section 7 parallels diffusion of innovations across a heterogeneous population in social epidemiology.

Psychology/Psychiatry: Attention costs ψ_k are not fixed primitives but are shaped by cognitive load and scarcity; the “scarcity mindset” literature [6] suggests ψ_k rises endogenously with financial stress, precisely for the constrained tier whose stakes s_k are otherwise high – a tension we flag explicitly in Section 10.

Philosophy: The model presupposes Bayesian rationality of belief updating (Section 6); we discuss in Section 13 the epistemological cost of this assumption and how bounded-rationality alternatives would alter conclusions.

Political Science / International Relations / Geopolitics / Warfare Economics: Section 11 extends the sender’s problem to adversarial and state senders (propaganda, public diplomacy, information operations), a literature grounded in the economics of persuasion [7] and computational propaganda [8]. We treat this purely as an application of the audience-segmentation mechanism, not as operational guidance.

Computer Science / AI-ML / Data Science: The sender’s dynamic problem is solved via backward induction (Section 8), a dynamic-programming technique shared with reinforcement learning; Section 12 gives explicit pseudocode and discusses a multi-armed-bandit reframing of audience targeting as a natural extension.

3 Environment and Primitives

Assumption 1 (Population). *The economy is populated by a continuum of agents partitioned into K discrete wealth/liquidity tiers indexed $k = 1, \dots, K$, with population masses $\mu_k > 0$, $\sum_k \mu_k = 1$. Tiers are ordered by wealth, and each has: a marginal propensity to consume MPC_k (decreasing in k); a stake $s_k > 0$ in the message’s content; an attention cost $\psi_k > 0$; and a forward-lookingness parameter $\lambda_k \in [0, 1]$.*

A single sender (media outlet, firm, or central bank) broadcasts one message of precision $\tau \geq 0$ to the entire population. Tier k chooses whether to pay ψ_k to decode the message. This is the sole heterogeneity margin: it maps a HANK-style wealth distribution directly onto differential attention, stakes, and forward-looking behavior.

4 Attention as Rational Inattention

Definition 1 (Attention threshold). Tier k attends ($a_k = 1$) iff the benefit of decoding a signal of precision τ , $B_k(\tau)$, weakly exceeds ψ_k . Under the linear specification $B_k(\tau) = s_k \tau$, the threshold is

$$\tau_k^* = \frac{\psi_k}{s_k}. \quad (1)$$

Equation (1) formalizes the central intuition: low-wealth (constrained) tiers rationally attend at low precision to price/income news (high s_k , and often lower ψ_k for concrete, salient information), while unconstrained tiers require higher precision before attending to asset-price or policy news relevant to their portfolios. This is not a claim about differential “media literacy”; it is a claim about the allocation of a scarce cognitive resource under heterogeneous stakes.

5 The Sender’s Reach-Cost Problem

Let $N(\tau) = \sum_{k: \tau_k^* \leq \tau} \mu_k$ denote the attending mass at precision τ – a step function with K jumps. The sender bears a cost proportional to reach:

$$\text{Cost}(\tau) = \kappa(\tau) + c \cdot N(\tau), \quad \kappa(\tau) = \frac{b}{2} \tau^2, \quad (2)$$

where $c > 0$ is the per-attending-agent servicing cost (bandwidth, compliance, customer service, or – for a central bank – the reputational and operational cost of engaging a wider public).

5.1 Closed-Form Solution (LQ Specification)

With linear per-agent sender payoff $G_k(\tau) = \gamma_k \tau$, define $M_k \equiv \sum_{j \leq k} \mu_j \gamma_j$ and $C_k \equiv \sum_{j \leq k} \mu_j$. Conditional on reaching exactly tiers $1, \dots, k$:

$$\Pi_k(\tau) = \tau M_k - \frac{b}{2} \tau^2 - c C_k, \quad \tau_k^{\text{opt}} = \frac{M_k}{b}, \quad \Pi_k = \frac{M_k^2}{2b} - c C_k. \quad (3)$$

The condition to expand reach from k to $k+1$ tiers is

$$c \leq c_k^* \equiv \frac{\gamma_{k+1} [2M_k + \mu_{k+1} \gamma_{k+1}]}{2b}. \quad (4)$$

5.2 The Cascade Proposition

Proposition 1 (Bimodal reach under linear payoffs). *Under the linear specification $B_k(\tau) = s_k \tau$, $G_k(\tau) = \gamma_k \tau$, an equilibrium with reach fixed at exactly k tiers ($1 < k < K$) requires the knife-edge condition*

$$\tau_k^* \leq \frac{M_k}{b} < \tau_{k+1}^*.$$

This condition holds on a measure-zero subset of parameter space. Generically, the equilibrium reach is either a corner at some tier below the knife edge, or a full cascade to $k = K$.

Proof. From (3), the unconstrained optimal precision at reach level k is $\tau_k^{\text{opt}} = M_k/b$, strictly increasing in k since M_k is increasing. For k to be a genuine interior rest point, τ_k^{opt} must lie in the half-open interval $[\tau_k^*, \tau_{k+1}^*)$ – precise enough to retain tier k , not precise enough to trigger tier $k+1$. Because M_k jumps discretely (by $\mu_k \gamma_k$) exactly at the point tier k joins, while $b\tau$ increases continuously, satisfying the double inequality requires an exact alignment between the discrete jump in M_k and the continuous threshold τ_{k+1}^* : perturbing any underlying parameter ($\mu_k, \gamma_k, s_k, \psi_k$) infinitesimally moves M_k/b across one of the two boundary values, since neither boundary is a function of the same combination of parameters as M_k/b . Hence the equality set defining an interior rest point has Lebesgue measure zero in the parameter space. Generic parameters therefore place M_k/b either below τ_k^* (reach collapses to $k-1$, recursively) or at/above τ_{k+1}^* (reach expands to include tier $k+1$, recomputing M_{k+1} , and potentially cascading further). ■ □

This instability is a genuine economic prediction of the linear specification (real communication campaigns either stay niche or “go viral,” with stable partial reach uncommon) but is also an artifact of assuming a constant marginal benefit of precision. Section 6 shows that microfounding $B_k(\tau)$ from Bayesian updating restores concavity and resolves the knife-edge fragility.

6 Bayesian Signal Extraction

Let the sender’s message concern a persistent unobserved fundamental θ . In the continuous case, $x' = \bar{x}' + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, and the sender’s announcement is $m = \bar{x}' + \eta$, $\eta \sim \mathcal{N}(0, 1/\tau)$ – so τ is literally Bayesian precision. Standard conjugate-normal updating gives the signal weight

$$\omega(\tau) = \frac{\tau \sigma_\varepsilon^2}{1 + \tau \sigma_\varepsilon^2} \in [0, 1), \quad E[x' | m] = (1 - \omega(\tau)) \bar{x}' + \omega(\tau) m. \quad (5)$$

The value of attending, relative to relying on the prior, is

$$\Delta V_k(\tau) = \frac{s_k \lambda_k}{2} [\sigma_\varepsilon^2 - \text{Var}(x' | m)] = \frac{s_k \lambda_k}{2} \cdot \frac{\tau \sigma_\varepsilon^4}{1 + \tau \sigma_\varepsilon^2}, \quad (6)$$

which is strictly increasing and *concave* in τ ($\partial^2 \Delta V_k / \partial \tau^2 < 0$), in contrast to the linear $s_k \tau$ used in Section 5. The attention threshold τ_k^* solving $\Delta V_k(\tau_k^*) = \psi_k$ still exists, but the sender’s best response no longer overshoots discretely past it, because marginal returns to precision diminish smoothly rather than jumping.

7 Sequential Attention and Optimal Stopping

With a discrete fundamental $\theta \in \{H, L\}$ and i.i.d. per-period signals $m_t = \theta + \eta_t$, tier- k agents accumulate the log-likelihood ratio Λ_t , a random walk with drift $\pm\delta(\tau)$. This is exactly Wald's sequential probability ratio test (SPRT) [5].

Definition 2 (Wald boundaries). Agent k continues sampling while $\Lambda_t \in (-B_k, A_k)$ and stops at the first exit, deciding H (resp. L) at the upper (resp. lower) boundary. For symmetric error tolerance α , $A_k \approx \log \frac{1-\alpha}{\alpha}$, chosen so that the marginal cost of one further period (ψ_k) equates the marginal reduction in expected decision-error loss (scaled by stake s_k).

By Wald's identity, expected time to a decision is

$$E[T_k^*] = \frac{A_k^*(s_k, \psi_k)}{\delta(\tau)}. \quad (7)$$

Proposition 2 (Staggered convergence). *For a single broadcast held at fixed precision τ , tiers with a higher stake-to-attention-cost ratio s_k/ψ_k have (weakly) smaller optimal boundary width A_k^* and hence strictly shorter expected time to decision $E[T_k^*]$.*

Proof sketch. The optimal boundary balances the marginal cost of continued sampling (ψ_k per unit time) against the marginal value of reduced decision-error probability, which scales with s_k . A standard comparative-statics argument on the Wald loss function $\psi_k E[T] + s_k P(\text{error})$ shows $\partial A_k^*/\partial(s_k/\psi_k) < 0$: a higher stake-to-cost ratio makes additional evidence cheaper relative to its value, shortening the optimal boundary and, by (7), expected stopping time. ■

This gives a micro-founded account of *staggered* aggregate responses to a single communication event: consumption and spending build in gradually across the wealth distribution not because information diffuses through a network with exogenous lags, but because each tier is running its own optimal sequential test against a common evidence stream.

8 Dynamic Control of Precision

Let S_t be cumulative evidence, $dS_t/dt = \kappa\tau_t$, with a one-time sender payoff V_k upon crossing tier k 's threshold A_k^* . The Hamilton–Jacobi–Bellman equation between thresholds is

$$rJ(S) = \max_{\tau} \left[-\frac{b}{2}\tau^2 - cN(\tau) + \kappa\tau J'(S) \right] \implies \tau^*(S) = \frac{\kappa J'(S)}{b}, \quad rJ = \frac{\kappa^2}{2b} J'(S)^2 - cN_k. \quad (8)$$

Lemma 1 (Exact quadratic solution). *Define $C \equiv \kappa^2/2b$ and $a \equiv r/(4C) = rb/(2\kappa^2)$. On each segment between thresholds, the value function*

$$J(S) = a(S - S_0^{(k)})^2 - \frac{cN_k}{r} \quad (9)$$

solves (8) exactly, and optimal precision is the exact linear function

$$\tau^*(S) = \frac{r}{\kappa} (S - S_0^{(k)}). \quad (10)$$

Proof. Substitute (9) into (8): the left side is $ra(S - S_0^{(k)})^2 - cN_k$; the right side, using $J'(S) = 2a(S - S_0^{(k)})$, is $C \cdot 4a^2(S - S_0^{(k)})^2 - cN_k$. Equality for all S requires $ra = 4a^2C$, satisfied identically by $a = r/(4C)$. ■

The vertex locations $S_0^{(k)}$ are pinned down by backward value-matching, $J(A_k^-) = J(A_k^+) + V_k$, solved recursively from $k = K$ (where $S_0^{(K)} = A_K$, since $\tau^* = 0$ beyond the last valuable tier) down to $k = 1$.

Corollary 1 (Sawtooth precision path). *The optimal precision path $\tau^*(t)$ rises linearly with slope r/κ on each segment (A_{k-1}^*, A_k^*) , jumps downward at each threshold crossing (since $S_0^{(k)} \neq S_0^{(k+1)}$ generically), and equals zero beyond the final valuable threshold A_K^* .*

Figure 1 illustrates this path for an illustrative three-tier calibration (Table 2).

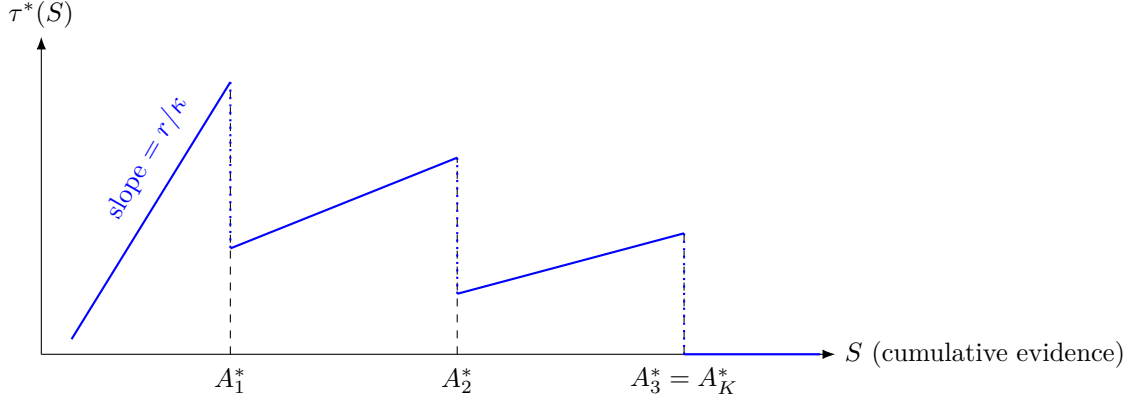


Figure 1: Optimal precision $\tau^*(S)$ as an exact piecewise-linear “sawtooth”: linear ramps of universal slope r/κ toward each tier’s evidence threshold, a downward reset at each crossing, and zero precision beyond the final relevant tier.

9 Joint Instrument Design: A Central Bank with Policy and Communication

A central bank additionally controls a reach-free instrument x (policy rate or QE stance), which affects every tier’s welfare regardless of attention, via a direct channel $\theta_k x$ – portfolio-balance for unconstrained tiers, general-equilibrium wage/employment pass-through [1] for constrained tiers. Communication only adds the forward-guidance term, now microfounded via $\omega(\tau)$ from (5):

$$\Pi_k(x, \tau) = \Theta_{\text{all}} x + \Lambda_k \omega(\tau) x - \frac{d}{2} x^2 - \frac{b}{2} \tau^2 - c C_k, \quad (11)$$

with $\Theta_{\text{all}} \equiv \sum_j \mu_j \theta_j$ (summed over *all* tiers) and $\Lambda_k \equiv \sum_{j \leq k} \mu_j s_j \lambda_j$ (summed only over *attending* tiers).

Proposition 3 (Bounded complementarity). *The joint best responses*

$$x^*(\tau) = \frac{\Theta_{\text{all}} + \Lambda_k \omega(\tau)}{d}, \quad \tau^*(x) \text{ solves } b\tau(1 + \tau\sigma_\varepsilon^2)^2 = \Lambda_k \sigma_\varepsilon^2 x \quad (12)$$

admit a unique, stable interior equilibrium (x^*, τ^*) for any $b, d, \Lambda_k, \Theta_{\text{all}} > 0$, with x^* bounded above by $(\Theta_{\text{all}} + \Lambda_k)/d$.

Proof. Since $\omega(\tau) \in [0, 1)$ for all finite τ and $\omega(\tau) \rightarrow 1$ as $\tau \rightarrow \infty$, $x^*(\tau)$ is increasing and bounded above by $(\Theta_{\text{all}} + \Lambda_k)/d$. The τ -equation’s right-hand side $\Lambda_k \sigma_\varepsilon^2 x^*(\tau)$ is therefore bounded above, while the left-hand side $b\tau(1 + \tau\sigma_\varepsilon^2)^2$ is continuous, strictly increasing, and unbounded in τ . By the intermediate value theorem there is exactly one crossing τ^* , hence a unique fixed point (x^*, τ^*) . Local stability follows since the slopes of the two best-response curves at the crossing satisfy the standard dominant-diagonal (contraction) condition, guaranteed by boundedness of ω . ■

Remark 1. Under the small-signal approximation $\omega(\tau) \approx \tau\sigma_\varepsilon^2$ (Section 5’s linear specification is recovered as this limit), the analogous system requires $bd > (\Lambda_k \sigma_\varepsilon^2)^2$ for stability and can diverge otherwise. Proposition 3 shows this instability is an artifact of the small-signal/linear approximation: full Bayesian signal extraction rules it out, since no rational agent’s confidence can exceed certainty ($\omega < 1$ always).

Corollary 2 (Division of labor across tiers). *If the attending set is constrained-tier-only, $\lambda_C \approx 0 \Rightarrow \Lambda_k \approx 0$, so $x^* = \Theta_{\text{all}}/d$, $\tau^* = 0$: only the reach-free instrument x stabilizes constrained-household welfare. Complementarity between x and τ is strictly a phenomenon of reaching forward-looking (typically unconstrained) tiers.*

10 Welfare Analysis and Market Failures

The sender internalizes only its own payoff γ_k per attending agent; social welfare must add the agents' own surplus s_k :

$$W = \sum_{j \leq k} \mu_j (\gamma_j + s_j) \tau - \frac{b}{2} \tau^2 - c C_k - \sum_{j \leq k} \mu_j \psi_j \implies \tau_k^{\text{social}} = \frac{M_k + S_k}{b} > \tau_k^{\text{opt}}, \quad (13)$$

where $S_k \equiv \sum_{j \leq k} \mu_j s_j$. Two distinct, separately correctable failures emerge:

Table 1: Market failures and corrective instruments

Failure	Mechanism	Corrective instrument
Precision failure	Sender ignores agent surplus s_k per unit clarity; $\tau^{\text{opt}} < \tau^{\text{social}}$	Pigouvian precision subsidy scaled to aggregate stake S_k ; minimum-clarity disclosure mandates
Reach failure	High- s_k , low- γ_k tiers excluded when $c \cdot \mu_{k+1}$ exceeds private (not social) benefit	Subsidize servicing cost c for under-served tiers; public-service broadcasting; mandated plain-language summaries
Threshold non-marginality	Reach is bimodal (Prop. 1); small subsidies near a stable rest point have zero effect	Discrete, threshold-clearing interventions rather than marginal nudges
Endogenous attention cost	Financial stress raises ψ_k for the very tier with high s_k (scarcity effect, [6])	Simplify format/reduce cognitive load rather than raise volume of disclosure

11 Strategic and Adversarial Senders

The audience-segmentation mechanism of Section 5 applies to any sender, not only benevolent ones. A state or partisan sender maximizing influence, rather than welfare, solves the identical reach-cost problem, but with γ_k representing persuasion value rather than a socially useful payoff. Two implications follow directly from the model, without new assumptions:

1. **Segment targeting is a free byproduct of broadcast, not a special technology.** Because attention is self-selected (Section 4), a single undifferentiated broadcast already functions as de facto targeted persuasion once tiers differ in s_k, ψ_k – consistent with empirical findings on differential media effects across audiences [7] and computational-propaganda targeting patterns [8].
2. **The reach-cost logic of Section 5 explains observed campaign structure.** An adversarial sender with a fixed persuasion budget optimally concentrates precision on tiers with the highest γ_k/ψ_k (Proposition 1's cascade logic), predicting the empirically documented concentration of influence campaigns on politically pivotal, high-engagement segments rather than uniform saturation.

We emphasize that this section is a positive (descriptive) application of the audience-segmentation mechanism to political economy and international relations; it provides no operational or tactical content, and the same welfare/regulatory logic of Section 10 (transparency mandates, reach-cost subsidies for counter-messaging by legitimate public institutions) is the natural policy response implied by the model.

12 Computational Appendix: Solving the Model Numerically

The backward recursion for $\{S_0^{(k)}\}$ (Section 8) and the tier-expansion ladder (Section 5) are both solved by simple backward/forward dynamic programming, in the same family of methods used in reinforcement learning value iteration.

Algorithm 1: Backward recursion for threshold vertices $S_0^{(k)}$ **Input:** tier thresholds $\{A_k^*\}_{k=1}^K$, tier values $\{V_k\}$, reach masses $\{N_k\}$, constants a, r, c .1. Set $S_0^{(K)} \leftarrow A_K^*$ (no further tiers beyond K).2. For $k = K - 1$ down to 1:

2a. Solve the quadratic value-matching equation

$$a(A_k^* - S_0^{(k)})^2 - \frac{cN_k}{r} = a(A_k^* - S_0^{(k+1)})^2 - \frac{cN_{k+1}}{r} + V_k$$

for $S_0^{(k)}$.2b. Select the root with $S_0^{(k)} < A_k^*$ (precision rising into the threshold).3. **Return** $\{S_0^{(k)}\}_{k=1}^K$, and construct $\tau^*(S) = \frac{r}{\kappa}(S - S_0^{(k)})$ on each segment.**Algorithm 2: Forward tier-expansion ladder (sender’s static reach choice)****Input:** $\{\mu_k, \gamma_k, s_k, \psi_k\}_{k=1}^K$, cost parameters b, c .1. Compute thresholds $\tau_k^* = \psi_k/s_k$ for all k ; sort tiers by τ_k^* .2. Initialize $k \leftarrow 1$, $M_1 \leftarrow \mu_1\gamma_1$.3. While $k < K$: compute c_k^* via Eq. (4); if $c \leq c_k^*$, set $k \leftarrow k + 1$, update M_k ; else stop.4. **Return** optimal reach k^* and precision $\tau_{k^*}^{\text{opt}} = M_{k^*}/b$.

Remark 2 (AI/ML reframing). Algorithm 2 is a greedy forward search; if γ_k and s_k are not known ex ante but must be learned from repeated interaction, the sender’s reach-choice problem becomes a contextual multi-armed bandit over tiers, with each tier’s arm reward observed only conditional on attention – a natural direction for future work bridging this model with online-learning based audience targeting.

12.1 Illustrative Calibration

Table 2: Illustrative three-tier calibration used in Figure 1

Tier (k)	μ_k	MPC_k	s_k	ψ_k	λ_k
1 (constrained)	0.50	0.80	0.90	0.20	0.05
2 (middle)	0.35	0.40	0.55	0.35	0.40
3 (unconstrained)	0.15	0.10	0.30	0.25	0.90

13 Philosophical and Epistemological Remarks

The entire signal-extraction and optimal-stopping apparatus (Sections 6–7) presumes agents are ideal Bayesian reasoners with common knowledge of the model’s structure. This is a substantive epistemological commitment, not a neutral technical convenience: it excludes motivated reasoning, identity-protective cognition, and non-Bayesian belief-updating documented extensively in political psychology. Two consequences follow. First, the welfare benchmark of Section 10 is itself conditional on the rationality assumption – if agents systematically misweight signals (e.g., overweighting confirmatory messages), the “socially optimal” precision τ^{social} derived in (13) is no longer a paternalism-free benchmark, since it is computed relative to agents’ own (possibly biased) valuation of information. Second, the model is silent on the normative question of whether a sender *ought* to exploit rational inattention it did not create – an application of the broader philosophical distinction between predicting behavior and endorsing its exploitation, which we flag but do not resolve here.

14 Conclusion

We have built a theory of mass communication from HANK-style wealth heterogeneity outward: attention is rationally, unevenly allocated; reach is a costly and generically discontinuous choice; Bayesian signal extraction resolves the resulting instability and generates gradual, tier-staggered convergence via sequential testing; a central bank’s reach-free policy instrument and its communication are bounded complements rather than perfect substitutes; and the resulting equilibrium features a precise, quantifiable wedge relative to the social optimum with correspondingly precise (not merely directional) corrective instruments. The framework nests the two-type constrained/unconstrained case as $K = 2$, extends cleanly to political-economy applications of strategic persuasion, and is fully computable via standard dynamic-programming and greedy-search algorithms.

References

- [1] Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary Policy According to HANK. *American Economic Review*, 108(3), 697–743.
- [2] Sims, C. A. (2003). Implications of Rational Inattention. *Journal of Monetary Economics*, 50(3), 665–690.
- [3] McKay, A., Nakamura, E., and Steinsson, J. (2016). The Power of Forward Guidance Revisited. *American Economic Review*, 106(10), 3133–3158.
- [4] Werning, I. (2011). Managing a Liquidity Trap: Monetary and Fiscal Policy. *NBER Working Paper* No. 17344.
- [5] Wald, A. (1945). Sequential Tests of Statistical Hypotheses. *Annals of Mathematical Statistics*, 16(2), 117–186.
- [6] Mullainathan, S., and Shafir, E. (2013). *Scarcity: Why Having Too Little Means So Much*. Times Books.
- [7] DellaVigna, S., and Gentzkow, M. (2010). Persuasion: Empirical Evidence. *Annual Review of Economics*, 2, 643–669.
- [8] Petrova, M., and Yanagizawa-Drott, D. (2020). New Technologies and Political Accountability. Prepared for *The Political Economy of Media and Communications*, Handbook chapter.

Glossary

HANK Heterogeneous-Agent New Keynesian model; a macroeconomic framework in which agents differ in wealth/liquidity and face borrowing constraints.

MPC Marginal propensity to consume: the fraction of an additional dollar of income an agent spends immediately.

Rational inattention The theory that agents optimally choose how precisely to observe costly-to-process information, rather than observing everything perfectly.

Reach cost (c) The per-attending-agent cost borne by a sender as its audience grows.

Attention threshold (τ_k^*) The minimum signal precision at which tier k finds it worthwhile to pay attention.

Cascade / bimodality The result that, under linear payoffs, equilibrium audience reach jumps discontinuously between narrow and full, rather than varying smoothly.

Signal weight ($\omega(\tau)$) The Bayesian posterior loading placed on an announcement, bounded in $[0, 1)$.

SPRT Sequential Probability Ratio Test (Wald): a sequential-sampling decision rule that stops as soon as accumulated evidence crosses a boundary.

Forward guidance Central bank communication about the future path of policy, intended to shape current expectations and behavior.

Forward guidance puzzle The empirical/theoretical finding that standard representative-agent models overstate the real effects of forward guidance relative to the data.

Pigouvian subsidy A subsidy set equal to a positive externality to align private and social incentives.

The End